An explicit invariant for a degree five symbol descent family

October 13, 2026

Abstract

Let K be a characteristic-zero field containing a primitive fifth root of unity q. Consider a central division algebra with symbol generators U, V, relation $UV = qVU$, scalar fifth powers, and the actual 25-element symbol basis. Suppose it admits an order-three semilinear automorphism cycling $U, V, (UV)^{-1}$ and fixing q. We describe a 69-term Laurent expression N, with a short expression using ordered pairs, which is fixed and nonscalar in every such division-algebra specialization. An elementary interpolation argument proves that the fixed division algebra has degree five over the fixed scalar field. An exact finite identity $N^5 = H(U^5, V^5)$, with a displayed 75-term integer Laurent polynomial H, gives a cyclic maximal subfield generated by N. The construction is motivated by Fryer's invariant generators and uses no primitive cube root in the hypotheses. The complete fifth-power certificate and the relative-family theorem have passed Lean checking, with no extra power hypothesis in the final theorem. A staged replay has also passed, with earlier checked objects reused as detailed below. Historical novelty is unestablished. The argument concerns this explicit semilinear symbol family and does not settle unrestricted prime-degree cyclicity.

Verification boundary. The full evaluated fifth-power certificate and the final relative-family assembly have passed Lean checking. The final export assumes exactly the structural symbol data stated below, with no additional numerator or power hypothesis. Its audit reports only the standard logical principles `propext`, `Classical.choice`, and `Quot.sound`; the literal output alignment uses no axioms. The N4 arithmetic and structural rebuilds, fresh N5 checks, and downstream final-assembly checks have passed. The final staged replay reuses 99 original earlier base/N2/N3 objects; it is not a fresh rebuild of the entire arithmetic closure. The distinction between these checks, the mathematical argument, and the additional paper-level description of a Galois generator is recorded in Section 9. No novelty or priority claim is made.

1 The precise family and the conclusion

All division algebras and field homomorphisms are unital. We identify a field with its specified central image when this causes no ambiguity. Negative powers are powers of inverses. Products are written in their intended order; coefficients in the center may be moved past all factors, but arbitrary algebra elements may not.

Definition 1.1 (Symbol data with an order-three action)**.** The data consist of the following objects and hypotheses.

(i) A field K of characteristic zero, a division algebra A with a central embedding $K \hookrightarrow A$, and $\mathrm{Z}(A) = K$.

(ii) An element $q \in K$ of multiplicative order five and units $U, V \in A$ satisfying

$$UV = qVU. \tag{1}$$

(iii) Elements $x, y \in K$ such that $U^5 = x$ and $V^5 = y$, and the assertion that

$$\{U^i V^j : 0 \le i, j < 5\} \tag{2}$$

is a K-basis of A.

(iv) Ring automorphisms $\sigma\colon A \to A$ and $\tau\colon K \to K$ satisfying, for every $c \in K$ and $a \in A$,

$$\begin{aligned} \sigma(ca) = \tau(c)\sigma(a), \qquad \sigma^3 = \mathrm{id}, \\ \sigma(U) = V, \qquad \sigma(V) = (UV)^{-1}, \qquad \tau(q) = q. \end{aligned} \tag{3}$$

Put

$$W = (UV)^{-1}, \qquad z = (xy)^{-1}, \qquad F = K^\tau, \qquad D = A^\sigma. \tag{4}$$

The fifth powers x, y are nonzero because U, V are units. The basis hypothesis implies $\dim_K A = 25$. We will prove, rather than assume, $\dim_F D = 25$ and $\mathrm{Z}(D) = F$. Likewise, invariance, nonscalarity, and the central fifth power of the particular numerator are not part of the data. Although $\sigma^3 = \mathrm{id}$ allows order dividing three in its wording, the hypotheses force both scalar and algebra automorphisms to have exact order three; see Lemma 2.2 and Lemma 2.4.

For an intrinsic central division algebra with a ring automorphism σ, restriction to the center supplies τ, and multiplicativity supplies semilinearity. Definition 1.1 spells out these maps because the formal interface does not identify a field literally with a subset of a different type. It introduces no assumption about a general Galois descent theorem.

Family conclusion. For every datum in Definition 1.1, the explicitly specified $N \in A$ below generates a cyclic maximal subfield $F(N)$ of D/F, of degree five. The proof uses the exact Laurent identity in Proposition 6.1. The full conclusion is stated in Theorem 7.2.

This is a relative assertion about a supplied symbol presentation and a particular action. It does not assert that every degree-five division algebra admits such data, that every cubic descent of a cyclic algebra is cyclic, or that arbitrary scalar choices x, y give a division algebra. No construction of a generic Ore division ring is needed anywhere in the argument. Constructing a generic example is separate background and is not part of the current formal theorem.

2 Quantum monomials and the scalar cycle

We first record all commutation conventions, since a phase error changes the finite certificate.

Lemma 2.1 (Integer exponent rules)**.** *For integers i, j, k, l one has*

$$V^j U^k = q^{-jk} U^k V^j, \tag{5}$$

$$(U^i V^j)(U^k V^l) = q^{-jk} U^{i+k} V^{j+l}, \tag{6}$$

$$(U^i V^j)^n = q^{-ijn(n-1)/2} U^{in} V^{jn} \qquad (n \ge 0). \tag{7}$$

In particular, $(UV)^5 = xy$, $W^5 = z$, and

$$(UV^{-1})^5 = xy^{-1}. \tag{8}$$

Proof. Equation (1) gives $VU = q^{-1}UV$. Multiplying this equality on the appropriate side by inverses gives the corresponding relations with U^{-1} and V^{-1}. Induction first on a nonnegative exponent and then on its negative establishes (5) for all integers. Multiplying $U^i(V^jU^k)V^l$ gives (6).

For (7), the assertion is immediate when $n = 0$. Multiplication on the right by U^iV^j adds the phase q^{-ijn}, and $n(n-1)/2 + n = n(n+1)/2$. This proves the induction. With $n = 5$ the phase is $q^{-10ij} = 1$, because $q^5 = 1$. Thus $(UV)^5 = U^5V^5 = xy$, and taking inverses gives $W^5 = z$. The same calculation with $(i, j) = (1, -1)$ gives (8). □

Lemma 2.2 (The induced scalar action)**.** *The automorphism σ cycles U, V, W, and τ cycles x, y, z. Moreover $q \in F$, and it has order five in F.*

Proof. Since $W = V^{-1}U^{-1}$, we have $VW = U^{-1}$. Therefore $\sigma(W) = (\sigma(U)\sigma(V))^{-1} = (VW)^{-1} = U$. Applying semilinearity to $c \cdot 1$ shows $\sigma(c) = \tau(c)$ on the specified copy of K. Consequently

$$\tau(x) = \sigma(U^5) = V^5 = y, \qquad \tau(y) = \sigma(V^5) = W^5 = z, \qquad \tau(z) = \sigma(W^5) = U^5 = x.$$

Finally $\tau(q) = q$ places q in F. The inclusion $F \hookrightarrow K$ is injective, so it preserves the least positive exponent giving $q^n = 1$. □

Lemma 2.3 (A centrally split polynomial in a division ring)**.** *If $t \in A$ satisfies $t^5 = 1$, then $t = q^i$ for some $0 \le i < 5$.*

Proof. The elements $1, q, \dots, q^4$ are distinct, and $X^5 - 1 = \prod_{i=0}^{4}(X - q^i)$ in $K[X]$. Evaluation at t respects multiplication because t commutes with every coefficient. Thus $0 = \prod_{i=0}^{4}(t - q^i)$ in A. A division ring has no zero divisors: if $ab = 0$ and $a \neq 0$, multiplication by a^{-1} gives $b = 0$. Repeated use of this fact shows that one of the displayed factors is zero. □

Lemma 2.4 (Pairwise distinct scalar parameters)**.** *The elements x, y, z are pairwise distinct. In particular, $(x - y)(x - z) \neq 0$.*

Proof. Suppose $x = y$. By (8), $t = UV^{-1}$ has $t^5 = 1$. Lemma 2.3 makes t central, so $U = q^iV$ for some i. Then U and V commute. Comparing with (1) yields $(1 - q)VU = 0$, impossible since $VU \neq 0$ and $q \neq 1$. Hence $x \neq y$. Applying the injective map τ to this inequality and then applying it again gives $y \neq z$ and $z \neq x$. The nonzero product assertion follows in the field K. Restricting $\sigma^3 = \mathrm{id}$ to the embedded field gives $\tau^3 = \mathrm{id}$. Since $\tau(x) = y \neq x$, τ has exact order three, and the same is then true of σ. □

Remark 2.5. This proof uses the division hypothesis essentially. A noncentral matrix can satisfy $t^5 = 1$, so the same argument cannot be transferred to split matrix algebras just because their formal symbol relations look similar. The conclusions of this paper apply to division-algebra specializations satisfying Definition 1.1.

3 Elementary descent of dimension and center

The following interpolation calculation supplies exactly the descent statements needed here. It avoids placing the desired fixed-algebra properties in the hypotheses.

Proposition 3.1 (A fixed spanning formula)**.** *For $a \in A$, define*

$$t_0(a) = a + \sigma(a) + \sigma^2(a),$$
$$t_1(a) = xa + y\sigma(a) + z\sigma^2(a),$$
$$t_2(a) = x^2a + y^2\sigma(a) + z^2\sigma^2(a).$$

Each $t_i(a)$ belongs to D, and

$$a = \frac{t_2(a) - (y+z)t_1(a) + yz\, t_0(a)}{(x-y)(x-z)}. \tag{9}$$

Consequently D spans A over K.

Proof. Applying σ to $t_0(a)$ cyclically permutes its three summands. For $t_1(a)$, semilinearity and Lemma 2.2 give

$$\sigma(t_1(a)) = y\sigma(a) + z\sigma^2(a) + xa = t_1(a).$$

The calculation for $t_2(a)$ is identical with squared coefficients. Expand the numerator of (9), retaining the order of the algebra elements. Since x, y, z are central, the coefficients of $a, \sigma(a), \sigma^2(a)$ are respectively

$$x^2 - (y+z)x + yz = (x-y)(x-z), \quad y^2 - (y+z)y + yz = 0, \quad z^2 - (y+z)z + yz = 0.$$

The denominator is a nonzero central scalar by Lemma 2.4; division is therefore legitimate. Equation (9) expresses every a as a K-linear combination of three elements of D. □

Theorem 3.2 (Structure of the fixed algebra)**.** *Under Definition 1.1, D is a central division algebra over F with*

$$\dim_F D = 25, \qquad \mathrm{Z}(D) = F.$$

Thus D has degree five.

Proof. Fixed elements are closed under addition, subtraction, and multiplication. If $d \in D$ is nonzero, then $\sigma(d^{-1}) = \sigma(d)^{-1} = d^{-1}$, so D is a division ring. Likewise F is a field. Its elements lie in D and are central there.

By Proposition 3.1, a K-basis of A can be selected from D: begin with the empty independent family and repeatedly select a fixed element outside its K-span until the span is all of A. This terminates with 25 elements because (2) gives $\dim_K A = 25$. Write this selected basis as $d_1, \ldots, d_{25}$. It is F-linearly independent, because any relation with coefficients in F is also a relation with coefficients in K.

For any $d \in D$ there are unique $k_i \in K$ with $d = \sum_i k_i d_i$. Apply σ and use fixedness of d, d_i:

$$\sum_i k_i d_i = d = \sum_i \tau(k_i) d_i.$$

Uniqueness of coordinates in a K-basis gives $\tau(k_i) = k_i$ for every i. Hence all k_i lie in F, and the selected family is an F-basis of D. This proves the dimension assertion.

Now let $c \in \mathrm{Z}(D)$. It commutes with each $t_i(a)$ in (9) and with every scalar in K, because $K = \mathrm{Z}(A)$. That equation implies $ca = ac$ for all $a \in A$. Therefore $c \in \mathrm{Z}(A) = K$, and its fixedness gives $c \in F$. The reverse inclusion $F \subseteq \mathrm{Z}(D)$ was already noted. A finite-dimensional central division algebra has degree equal to the square root of its central dimension; here that degree is five. □

The same selected basis shows that the multiplication map $D \otimes_F K \to A$, $d \otimes k \mapsto kd$, is an isomorphism of K-algebras: it maps the K-basis $d_i \otimes 1$ to the K-basis d_i. This observation explains the term "descent", but neither a tensor-product construction nor the general Galois descent theorem is used to prove Theorem 3.2.

4 A compact numerator without cube roots

A short expression is useful for proving invariance. A separate expanded expression is useful for exact multiplication and basis coordinates. We specify both and identify the bridge between them.

4.1 Ordered pairs and a formal central variable

For pairs of elements of A, define

$$(a, b) \star (d, e) = (ad - be,\ ae + bd - be). \tag{10}$$

Addition and scalar multiplication are componentwise. This is the multiplication obtained by writing $a + \xi b$ in the ring

$$A[\xi]/(\xi^2 + \xi + 1),$$

where ξ is a formal central variable and the quotient is by the two-sided ideal generated by the central monic polynomial $\xi^2 + \xi + 1$. Monic division by $\xi^2 + \xi + 1$ gives a unique representative of degree at most one; the product is therefore associative even though A need not be commutative. This auxiliary ring can have zero divisors. We never assert that it is a field or a division algebra, and we take no inverses in it.

In this pair notation, multiplication by ξ^2 and ξ is given by

$$T(a, b) = (b - a, -a), \qquad S(a, b) = (-b, a - b). \tag{11}$$

These equalities follow directly from $\xi^2 = -1 - \xi$ and $\xi^3 = 1$. Both operations commute with multiplication by central scalars. The map σ acts componentwise and respects $\star$, since the product only uses sums, differences, and products in A.

4.2 Definition of the numerator

Set

$$\mathcal{A} = (U - W,\ V - W), \qquad \mathcal{B} = (U^{-1} - W^{-1},\ V^{-1} - W^{-1}), \tag{12}$$

$$c = 5(1 + q^{-1}), \qquad (d_0, d_1) = (x - y,\ z - y), \tag{13}$$

$$(P_0, P_1) = (\mathcal{A}^{\star 4} - c\mathcal{A}) \star \mathcal{B}. \tag{14}$$

Here $\mathcal{A}^{\star 4} = (\mathcal{A} \star \mathcal{A}) \star (\mathcal{A} \star \mathcal{A})$, and c multiplies both components on the left. Define

$$N_{\mathrm{cmp}} = (2d_0 - d_1)P_0 + (2d_1 - d_0)P_1. \tag{15}$$

Every inverse in this expression is the inverse of one of the specified units or of q. In particular, the formula has no denominator involving $x - y$, a discriminant, or a sum of scalar monomials. It is defined for every datum in Definition 1.1.

Proposition 4.1 (Invariance of the compact formula)**.** *One has* $\sigma(N_{\mathrm{cmp}}) = N_{\mathrm{cmp}}$.

Proof. Using the cycle $U \mapsto V \mapsto W \mapsto U$,

$$\sigma(\mathcal{A}) = (V - U, W - U) = T\mathcal{A}.$$

The same calculation for the inverses gives $\sigma(\mathcal{B}) = T\mathcal{B}$. In the formal-variable notation, these are multiplication by ξ^2. Since c is fixed, ξ is central, and $\xi^8 = \xi^2$, we obtain

$$\sigma(\mathcal{A}^{\star 4} - c\mathcal{A}) = \xi^2(\mathcal{A}^{\star 4} - c\mathcal{A}), \qquad \sigma(P_0, P_1) = \xi^4(P_0, P_1) = S(P_0, P_1).$$

This calculation involves no specialization of ξ to an actual cube root in K.

The scalar pair transforms in the same way:

$$\tau(d_0, d_1) = (y - z, x - z) = (-d_1, d_0 - d_1) = S(d_0, d_1).$$

For a central pair $d = (d_0, d_1)$ and an arbitrary pair $P = (P_0, P_1)$, write $L(d, P) = (2d_0 - d_1)P_0 + (2d_1 - d_0)P_1$. A direct calculation, keeping all coefficients on the left, gives

$$\begin{aligned} L(Sd, SP) &= (-d_0 - d_1)(-P_1) + (2d_0 - d_1)(P_0 - P_1) \\ &= (2d_0 - d_1)P_0 + (2d_1 - d_0)P_1 = L(d, P). \end{aligned}$$

Applying this identity to the two transformed pairs proves the assertion. □

4.3 The expanded numerator and the checked bridge

Let $\mathcal{S} \subset \mathbb{Z}^2$ be the 69 exponent pairs in Appendix A. For each $(i, j) \in \mathcal{S}$, the table gives an integer quadruple (a_0, a_1, a_2, a_3), defining $n_{ij}(q) = a_0 + a_1 q + a_2 q^2 + a_3 q^3$. Put

$$N = \sum_{(i,j)\in\mathcal{S}} n_{ij}(q) U^i V^j. \tag{16}$$

Proposition 4.2 (Identification of the two formulas)**.** *With the phase convention* (6)*, one has* $N = N_{\mathrm{cmp}}$*. Consequently* $N \in D$*.*

Finite expansion and its verification. Replace W by $q^{-1}U^{-1}V^{-1}$ and W^{-1} by UV in (12); the first replacement follows from $W = V^{-1}U^{-1}$ and (5). Replace x, y, z by $U^5, V^5, U^{-5}V^{-5}$ in (13). Expand each of the pair products using (10), multiply ordered monomials using (6), and reduce coefficients with

$$q^4 = -1 - q - q^2 - q^3. \tag{17}$$

The resulting coefficients are exactly those in Appendix A. This is a finite coefficient assertion, rather than a commutative simplification of the noncommuting variables. The accompanying replay script explicitly performs these replacements and compares all coefficients, including absent coefficients, with the table. The recorded formal compact-to-Laurent bridge also proves this equality after evaluation in the actual division algebra; see Section 9. Thus invariance from Proposition 4.1 applies to the specific expanded numerator, not merely to a different expression with the same name. □

Relation to the motivating construction. Fryer's order-three construction uses an auxiliary primitive cube root ω and an invariant expressed as $a^{-1}b$; see [Fry14, Proposition 3.3 and equations (3.2.1)–(3.2.2)]. Our pairs encode the expressions $a = U + \omega V + \omega^2 W$ and $b = U^{-1} + \omega V^{-1} + \omega^2 W^{-1}$ without requiring that root in K. For a pair $p = p_0 + \xi p_1$, the formal substitution $\xi \mapsto \xi^2$ gives $\overline{p} = \underline{(p_0 - p_1)} - \xi p_1$. The sum $p + \overline{p}$ is $2p_0 - p_1$. In particular, (15) is the formal trace of $(P_0 + \xi P_1)\overline{(d_0 + \xi d_1)}$. This describes the trace-like shape of the formula; the proof uses the explicit polynomial expression, not an assertion that inverses survive an extension or specialization. No denominator-clearing step is used to infer nonscalarity after specialization.

5 Uniform nonscalarity from two coordinates

A noncentral term in a generic Laurent polynomial need not remain a nonzero noncentral coordinate after specializing central powers. The next argument deals with those possible cancellations explicitly.

Lemma 5.1 (Residue coordinates)**.** *For $i = 5a + r$ and $j = 5b + s$, with $a, b \in \mathbb{Z}$ and $0 \le r, s < 5$,*

$$U^i V^j = x^a y^b U^r V^s. \tag{18}$$

For the numerator (16)*, the coordinates on UV^3 and U^2V^3 in the basis* (2) *are respectively*

$$C_{13} = \frac{q(x^2y + xy^2 - 2)}{x^2y^2}, \tag{19}$$

$$C_{23} = -\frac{q^3(2x^2y - xy^2 - 1)}{xy^2}. \tag{20}$$

Proof. The equality $U^{5a+r} = x^a U^r$ is valid for positive and negative a, and similarly for V. The extracted powers of x, y are central, so moving them to the left introduces no phase. This proves (18).

Exactly three terms of Appendix A have residues $(1, 3)$:

$$(-9, -7; -2q), \qquad (-4, 3; q), \qquad (1, -2; q).$$

Their combined coefficient is $-2qx^{-2}y^{-2} + qx^{-1} + qy^{-1}$, which becomes (19) after a common denominator is taken. Exactly three terms have residues $(2, 3)$:

$$(-3, -7; q^3), \qquad (2, 3; q^3), \qquad (7, -2; -2q^3).$$

Their coefficient is $q^3x^{-1}y^{-2} + q^3 - 2q^3xy^{-1}$, giving (20). The remaining table entries belong to other residue classes and cannot contribute to these coordinates. □

Proposition 5.2 (Nonscalarity in every division specialization)**.** *One has $N \notin K$. In particular $N \neq 0$ and, as an element of D, $N \notin F$.*

Proof. The basis vector indexed by $(0, 0)$ is 1. Therefore an element of K has zero coordinates on all other basis vectors. If $N \in K$, both (19) and (20) vanish. Their denominators and q are nonzero, so, on putting $A_0 = x^2y$ and $B_0 = xy^2$, we obtain

$$A_0 + B_0 = 2, \qquad 2A_0 - B_0 = 1.$$

Adding the two equations gives $3A_0 = 3$. Characteristic zero implies $A_0 = 1$, and the first equation then gives $B_0 = 1$. Hence $0 = A_0 - B_0 = xy(x - y)$. Since $xy \neq 0$, this forces $x = y$, contradicting Lemma 2.4. Thus $N \notin K$. Since $0 \in K$ and $F \subseteq K$, the final assertions follow. □

The proof requires an actual basis, not merely a spanning list of 25 monomials. It also explains why a proof using a single generic coefficient would be insufficient for the stated family. The two-coordinate contradiction is insensitive to algebraic relations among x, y, provided the given algebra is a division algebra with the required basis.

6 The exact fifth power certificate

6.1 An integral twisted Laurent algebra

Let

$$R = \mathbb{Z}[Q]/(Q^4 + Q^3 + Q^2 + Q + 1),$$

and write Q for the residue class. It satisfies $Q^5 = 1$ and is a unit. Every element of R is represented uniquely by an integer quadruple (a_0, a_1, a_2, a_3), since division by the monic polynomial gives a unique remainder of degree at most three. Introduce a free R-module with basis e_{ij} indexed by $(i, j) \in \mathbb{Z}^2$ and finite support. Define

$$e_{ij}e_{kl} = Q^{-jk}e_{i+k,j+l}. \tag{21}$$

This multiplication is associative. On multiplying three basis vectors, the exponents of Q from the two parenthesizations are

$$-jk - (j + l)m \quad \text{and} \quad -lm - j(k + m),$$

which are equal. Bilinearity extends associativity to finite sums. The unit is e_{00}.

Equation (17) follows from $q^5 = 1$ and $q \neq 1$: $(q - 1)(1 + q + q^2 + q^3 + q^4) = q^5 - 1 = 0$, so the second factor vanishes in K. Thus $Q \mapsto q$ defines a ring map $R \to K$. By Lemma 2.1, the map

$$\mathrm{ev}(e_{ij}) = U^i V^j \tag{22}$$

extends to a ring homomorphism from the twisted Laurent algebra to A. It need not be injective: relations among x, y may identify central Laurent expressions. Injectivity is unnecessary for transporting a universal identity. The actual symbol basis is used only when extracting coordinates for nonscalarity and dimension.

Write $\mathcal{N} = \sum n_{ij}(Q)e_{ij}$ for the formal table in Appendix A. Define $H(X, Y)$ by Appendix B:

$$H(X, Y) = \sum_{(a,b)} h_{ab}X^aY^b \in \mathbb{Z}[X^{\pm 1}, Y^{\pm 1}]. \tag{23}$$

There are 75 nonzero entries in that table, all ordinary integers.

Proposition 6.1 (The exact fifth-power identity)**.** *In the twisted Laurent algebra one has*

$$\mathcal{N}^5 = \sum_{(a,b)} h_{ab}e_{5a,5b}. \tag{24}$$

Thus every coefficient at an exponent outside $5\mathbb{Z}^2$ is zero, and the right-hand side is exactly the table in Appendix B.

Exact finite certificate. Apply the coefficient reduction and convolution specified below to the 69-term table. The complete products have respectively 237, 552, 969, and 75 nonzero terms. Every multiplication equality is established by finite exact arithmetic; the last output is coefficientwise equal to the 75-term table, with exponents multiplied by five. These equalities, not merely the support counts, prove (24) using (21).

The checked proof package supplies all multiplication certificates in the full fifth-power chain, a soundness theorem transporting their operations to evaluated products, and a kernel-checked

equality identifying the final literal output with the standalone 75-term candidate list. The checked support theorem and integer-coefficient theorem apply to that same list. The complete evaluated theorem is `PrimeCyclicity.N5Modern.numerator_fifth`; final alignment is `PrimeFiveFamily.fifthOutput_eq_candidate`. The latter has no axioms. The exact recurrence below and the complete input and output tables allow independent arithmetic replay. No assertion of equality is inferred solely from a central-looking output list or a numerical specialization. □

Status of the certificate. The first full Lean check is complete, including semantic evaluation and final-family integration. Recorded external exact-arithmetic computations and the standalone script supplied with this paper also reproduce the identity. The independent N4 and structural rebuilds, fresh N5 replay, and downstream final-assembly checks have passed. Original earlier base/N2/N3 objects are reused, so this is not a fresh rebuild of the entire fifth-power dependency closure. The Lean export is the universal evaluated identity in its specified algebraic interface. The twisted Laurent algebra here is the mathematical presentation of the same finite arithmetic, not a claim that Lean separately implements this particular quotient-and-free-module definition.

6.2 The finite computation in reproducible form

For clarity, the full arithmetic algorithm is specified independently of a particular programming language. Represent a Laurent expression as a finite map $C\colon \mathbb{Z}^2 \to \mathbb{Z}^4$, with omitted entries interpreted as zero. For coefficient quadruples a, b and an integer s, let

$$\mathrm{cmul}(a, b; s) = \mathrm{red}\big(Q^{\bar{s}}(a_0 + a_1Q + a_2Q^2 + a_3Q^3)(b_0 + b_1Q + b_2Q^2 + b_3Q^3)\big),$$

where $\bar{s} \in \{0, 1, 2, 3, 4\}$ is congruent to s modulo five and red is the monic remainder modulo $1 + Q + Q^2 + Q^3 + Q^4$. To compute the latter, process the highest nonzero degree $d \geq 4$, remove its coefficient tQ^d, and subtract t from the coefficients at degrees $d-4, d-3, d-2, d-1$. This replaces Q^d by $-Q^{d-4} - Q^{d-3} - Q^{d-2} - Q^{d-1}$ and strictly lowers the largest degree still requiring reduction. It terminates and gives the unique remainder.

The product of maps C, E is

$$(C * E)_{r,s} = \sum_{\substack{i+k=r \\ j+l=s}} \mathrm{cmul}(C_{i,j}, E_{k,l}; -jk). \tag{25}$$

Only pairs in the two finite supports enter the sum. Add quadruples componentwise and delete entries that become the zero quadruple. Starting with $C^{(1)} = (n_{ij})$, form $C^{(m+1)} = C^{(m)} * C^{(1)}$ for $m = 1, 2, 3, 4$. An exhaustive check compares $C^{(5)}$ with the right-hand map in (24). The comparison must be equality of finite maps, not agreement on a selection of positions: otherwise noncentral residual terms could be missed.

The recorded nonzero support sizes are

$$69, \quad 237, \quad 552, \quad 969, \quad 75$$

for powers one through five. The first four maps have no support in $5\mathbb{Z}^2$; all support of the fifth map is in $5\mathbb{Z}^2$. Those counts are useful diagnostics but do not prove equality. Many different maps have the same support size. Only coefficientwise equality, with all omitted coefficients checked to be zero, supplies the identity.

6.3 Why the arithmetic gives centrality after specialization

Lemma 6.2 (Semantic consequence of the certificate)**.** *One has*

$$N^5 = H(x, y) \in K.$$

Moreover $H(x, y) \in F^\times$.

Proof. Apply the ring homomorphism (22) to (24). The left side becomes N^5, not merely the evaluation of an unrelated output list. For each right-hand term, $U^{5a}V^{5b} = x^a y^b$ by (18); the result is $H(x, y)$.

Proposition 4.2 gives $\sigma(N) = N$. Hence

$$\tau(H(x, y)) = \sigma(H(x, y)) = \sigma(N^5) = \sigma(N)^5 = N^5 = H(x, y).$$

The equality is interpreted in K using its injective embedding in A, so $H(x, y) \in F$. By Proposition 5.2, $N \neq 0$. A nonzero element of a division ring has nonzero fifth power. Therefore $H(x, y) \neq 0$. □

This last nonvanishing conclusion does not require a separate analysis of the zeros of the 75-term polynomial. It is a consequence of the exact identity and uniform nonscalarity in the given division algebra.

6.4 Arithmetic certificates versus a trusted computation

A generator may compute candidate lists and place them in Lean source, but that generator is not a logical oracle. The checking layer proves finite multiplication equalities by kernel reduction of exact integer and list operations. Splitting a product into rows or smaller pieces changes the proof's size, not the required equality. Each row has to be joined to the exact partial sum from the preceding rows, and the exported theorem has to cover the whole input list.

A second layer proves that the finite operations agree with evaluation in an arbitrary division algebra satisfying the scalar and commutation hypotheses. A closed finite-list certificate alone does not prove that its output evaluates to N^5. Conversely, a generic evaluation-multiplication lemma alone does not establish that the specific output has central support. The final result needs both layers and the link to the frozen numerator.

For the completed cube stage, the recorded audit reports no axioms for the closed arithmetic certificates and only the standard logical principles `propext`, `Classical.choice`, and `Quot.sound` for the semantic theorem. These reports concern their named declarations and imports. The separate full fifth-power and final-assembly audits now supply the corresponding checked coverage for those completed stages. Hashes identify a snapshot; they do not independently prove a mathematical statement.

7 The cyclic maximal subfield endpoint

We give the field argument explicitly. In this section, "cyclic division algebra" means a central division algebra containing a maximal subfield that is a cyclic Galois extension of its center. This is the usual maximal-subfield characterization; no presentation by a second cyclic algebra generator is needed for the conclusion stated here.

Lemma 7.1 (A fifth-central element in central dimension twenty five)**.** *Let B be a division algebra with center E, $\dim_E B = 25$, and $\operatorname{char} E = 0$. Suppose $q \in E$ has order five and $r \in B$ satisfies $r \notin E$ and $r^5 = h \in E$. Then $E[r] = E(r)$ is a cyclic Galois extension of E of degree five and is a maximal commutative subfield of B.*

Proof. Because E is central, the subring $E[r]$ is commutative. The equation $r^5 = h$ shows that it is spanned over E by $1, r, r^2, r^3, r^4$. It is a domain, being a subring of a division ring. Multiplication by any nonzero element $s \in E[r]$ is an injective linear endomorphism of this finite-dimensional E-space and hence is surjective. In particular some $t \in E[r]$ satisfies $st = 1$. Thus $L = E[r]$ is a field, and equals $E(r)$.

Write $d = [L : E]$. The preceding spanning argument gives $d \leq 5$. Regard B as a left vector space over L. The scalar action need not be central for this left vector-space structure: multiplication in B is associative. A finite L-basis exists because an E-spanning set is also an L-spanning set. If $\ell_1, \ldots, \ell_d$ is an E-basis of L and $b_1, \ldots, b_m$ is a left L-basis of B, the products $\ell_i b_j$ form an E-basis of B. Spanning follows by expanding each left coefficient, and independence follows first from the L-independence of the b_j and then from the E-independence of the ℓ_i. Consequently $25 = dm$, so d divides 25. Since $r \notin E$, $d > 1$; together with $d \leq 5$ this forces $d = 5$.

The monic minimal polynomial of r over E divides $X^5 - h$ and has its full degree five. Hence it equals $X^5 - h$. Also $r \neq 0$, since $0 \in E$, so $h \neq 0$. The five elements $q^i r$ belong to L, are pairwise distinct, and each has fifth power h. Thus $X^5 - h$ splits into distinct linear factors in $L[X]$. Its irreducibility and the fact that L is its splitting field show that L/E is Galois. More explicitly, substitution $r \mapsto qr$ induces an E-automorphism of $E[X]/(X^5 - h) \simeq L$: it respects the relation and has inverse $r \mapsto q^{-1}r$. Its successive powers send r to $q^i r$ and have order exactly five. Since a degree-five extension has at most five E-automorphisms, this automorphism generates $\operatorname{Gal}(L/E)$.

Finally, let M be any commutative E-subalgebra of B containing L. It is finite dimensional and a domain, so the same multiplication argument makes it a field. The vector-space argument just given shows $[M : E]$ divides 25, and the field tower through L shows it is a multiple of five. Therefore $[M : E]$ is 5 or 25. In the former case $M = L$. In the latter case the E-subspace $M \subseteq B$ has the full dimension of B, hence $M = B$; this would make B commutative, contradicting $\mathrm{Z}(B) = E$ and $\dim_E B = 25 > 1$. Thus $M = L$, proving maximality. □

Theorem 7.2 (Explicit cyclicity for the specified family)**.** *Let the data satisfy Definition 1.1, and let N be the exact numerator of Appendix A, equivalently* (15)*. Then D/F is a central division algebra of degree five, and*

$$L = F(N) \subseteq D$$

is a cyclic maximal subfield of degree five. Its minimal polynomial is $X^5 - H(x, y)$ and a generator of $\operatorname{Gal}(L/F)$ sends N to qN.

Proof. Theorem 3.2 gives the central dimension and center of D. Propositions 4.2 and 5.2 put N in D but outside F. Proposition 6.1 and Lemma 6.2 give $N^5 = H(x, y) \in F^\times$. Lemma 2.2 places a primitive fifth root q in F. Apply Lemma 7.1 with $(B, E, r, h) = (D, F, N, H(x, y))$. □

Scope of the checked final export. The final export states degree five, Galoisness, cyclicity of the Galois group, and maximality of the generated commutative subalgebra, with no additional scalar fifth-power hypothesis. It obtains the required identity from the completed power certificate and applies the earlier conditional endpoint internally. The identification of the minimal polynomial as $X^5 - H(x, y)$ and the designation of the generator $N \mapsto qN$ are proved in the paper argument

above. They are not additional conclusions directly named in that final export. The relative-family structural assumptions remain essential; removing the intermediate power hypothesis does not remove any of those assumptions.

8 Relationship with the literature and limits of the claim

The motivation is an existing fixed-ring construction. Fryer studies the order-three monomial action on a quantum division ring and constructs fixed quantum generators under an auxiliary primitive cube-root hypothesis [Fry14, Proposition 3.3 and Theorem 3.4]. At a fifth root q, roots of q needed there can be chosen among powers of q, but a primitive cube root of unity need not be in the ground field. The present construction addresses a cyclicity conclusion for the restricted data in Definition 1.1, including division-algebra specializations; it does not reproduce a full fixed-ring isomorphism theorem. Fryer's Questions 4.12 do not explicitly formulate removal of the cube-root hypothesis as the problem treated here. No such open-problem attribution is made.

Reichstein and Vonessen discuss the related monomial action within an action of $(C_p)^2 \rtimes C_3$ on a generic symbol algebra [RV06, Section 15, especially Proposition 15.3]. Their obstruction concerns a maximal subfield invariant under the entire larger group. The present construction seeks a field in the C_3-fixed algebra, with no requirement of invariance under the additional $(C_5)^2$. These are different conditions. On exponent pairs the C_3 action has matrix

$$\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix},$$

whose characteristic polynomial $T^2 + T + 1$ has no root in $\mathbb{F}_5$. This simple calculation explains why an invariant monomial line is not the natural source of the nonmonomial candidate N.

Matzri's theorem proves cyclicity for degree-five division algebras with a specified dihedral splitting field [Mat08, Theorem 3.2]. The order-three semilinear action in our data does not supply that hypothesis. In particular, an argument that cyclicity holds after adjoining a cube root of unity cannot be replaced by an unjustified assertion that cyclicity descends through every quadratic extension. The distinction between cyclicity questions and particular splitting-field structures is also emphasized in the survey [ABGV11, Section 2]. That survey supplies historical context, not a claim about the present status of every unrestricted cyclicity problem.

A bounded targeted review of primary literature did not identify the exact root-free specialization-uniform statement or the exact numerator presented here. This negative search result does not establish originality. The family theorem, the explicit invariant, and its machine-checkable certificate would need separate comparison with the literature before any claim of novelty or priority. In particular, this paper is not described as a solution to an explicitly published cube-root-removal question.

9 Formal coverage and verification limits

The mathematical data correspond to `PrimeFiveFamily.SymbolFamily`. Its fields specify a characteristic-zero scalar field, a division algebra, a primitive fifth root, actual units, the quantum relation, scalar fifth powers, the actual monomial basis, the center characterization, and the two compatible automorphisms. The structure does not contain the fixed dimension, the fixed center, cyclicity, or any assertion about the numerator as input fields.

9.1 Structural and numerator interfaces

The structural family manifest records 26 audited declarations. An independent structural rebuild checked all 34 modules in the structural dependency chain from fresh project objects, with zero reused project objects. Official upstream Mathlib and dependency objects were reused; the receipt does not claim those libraries were rebuilt. This earlier rebuild ended at the explicitly conditional supporting endpoint. It is separate from the subsequent complete power and final-assembly checks.

The named interfaces make the connection to the mathematical argument precise.

- `frozen_eq_compact` identifies the evaluated frozen 69-term list with the compact expression, accounting for the actual U, V and commutation phase.
- `frozenNumerator_fixed` and `frozenNumerator_not_scalar` establish invariance and non-scalarity of that exact list. The definition `fixedNumerator` places it in the actual fixed algebra.
- `tau_x`, `tau_y`, `tau_z`, and `x_ne_y` derive the scalar cycle and inequality. The names `fixed_finrank`, `fixedDivisionRing`, and `fixed_center` record the elementary descent consequences.
- The earlier `supporting_cyclicity_of_fifthPower_identity` retains an explicit identity hypothesis. It remains a useful conditional lemma, but the final assembly now supplies that identity as a theorem.

9.2 Completed fifth power and final assembly

The complete evaluated fifth-power export is `PrimeCyclicity.N5Modern.numerator_fifth`, in `modern.N5Fifth`. Its literal output is `PrimeCyclicity.N5Modern.acc69`, with 75 terms. The recorded fifth-power audit covers 416 chain modules and reports matching source and main-object hashes. Closed arithmetic certificates have no axioms; the semantic theorem reports only `propext`, `Classical.choice`, and `Quot.sound`.

The separate checked helper `FrozenCentralOutput75.candidateOutput75_support` proves that every exponent in the expected output is divisible by five; its audit reports only `propext`. Its length and integer-coefficient assertions have no axioms. Crucially, final assembly proves `fifthOutput_eq_candidate`, identifying the completed literal output with that exact helper list by kernel computation and without axioms. The support certificate therefore applies to the actual power output, not merely a proposed table.

Next, `frozenNumerator_fifth` instantiates the universal evaluated identity with the actual scalar unit q and actual units U, V, then applies the central-support evaluator using $U^5 = x$ and $V^5 = y$. The scalar is defined by `fifthScalar`, the central coefficient evaluation of the matched 75-term list. Its mathematical value is $H(x, y)$ by the exponent normalization in Appendix B.

The final theorem is

`PrimeFiveFamily.symbolFamily_cyclicity`.

It proves degree five, Galoisness, cyclicity of the Galois group, and maximality among commutative subalgebras for the field generated by the actual fixed numerator. Its only data are the specified `SymbolFamily` and ambient field, characteristic-zero, division-ring, and algebra instances. There is no extra power, fixedness, nonscalarity, dimension, or cyclicity hypothesis. The final assembly and audit passed Lean 4.32.2 against the pinned Mathlib 4.32.2 environment using existing dependencies. Their final axiom report is exactly `propext`, `Classical.choice`, and `Quot.sound`.

The initial check and a staged replay of the relative-family theorem have passed. The final replay freshly compiled 418 N5 modules, a separate N5 audit, and `FrozenCentralOutput75`, `FinalAssembly`, and `FinalAudit`: 422 successful invocations, with no failures or retries. It reused 212 independently rebuilt N4 modules, 31 independently rebuilt structural modules, and 99 original earlier base/N2/N3 module objects. Of those 99, `modern.Algebra` and `modern.Numerator` also had separate source rebuilds; the remaining 97 retain original checked-object provenance. Source and object hashes were verified. This is not an independent fresh rebuild of all 763 custom sources. The included public verifier has not completed an end-to-end replay. Source files, receipts, statements, and hashes identify the checked artifact; a hash itself is not a proof.

9.3 Replay evidence and review limits

The terminal replay used 3348.363222 child CPU seconds. Its exact module inventory, per-source hashes, compiler receipts, final theorem statement, and axiom report accompany the source package. The replay used pinned upstream compiled objects; no whole-Mathlib source rebuild, separate-kernel implementation, fresh-network installation test, or external human peer review is claimed.

A bounded literature search does not establish originality. A specialist assessment would be needed before making a novelty or priority claim. The minimal-polynomial and designated-generator claims are mathematical arguments in this paper, distinct from the final Lean export. No author or runtime model identity is asserted by the paper.

The scope remains Definition 1.1. No generic Ore division ring has been constructed by the formal theorem, no positive-characteristic extension is asserted, and no unrestricted degree-five or prime-degree cyclicity problem is solved.

A The complete numerator table

The following table specifies N without reliance on a hidden coefficient file. Each row $(i, j; a_0, a_1, a_2, a_3)$ contributes $(a_0 + a_1q + a_2q^2 + a_3q^3)U^iV^j$ to (16). All omitted coefficients are zero. The order of U and V is part of the definition.

i	j	a_0	a_1	a_2	a_3
−10	−9	0	−1	0	0
−9	−10	−1	0	0	0
−9	−7	0	−2	0	0
−8	−8	−1	−2	−2	−1
−8	−5	1	1	1	1
−7	−9	0	0	0	−2
−7	−6	0	−2	−1	−1
−7	−3	1	0	0	0
−6	−7	−1	1	1	0
−6	−4	−1	−1	1	0
−6	−1	−2	−2	−2	−2
−5	−8	1	1	1	1
−5	−2	−2	−3	−1	−1
−5	1	0	2	0	0

Continued on the next page

i	j	a_0	a_1	a_2	a_3
−4	−6	0	1	1	2
−4	−3	−2	−2	−1	0
−4	0	−1	−1	0	0
−4	3	0	1	0	0
−3	−7	0	0	0	1
−3	−4	0	−1	−2	−2
−3	−1	0	0	0	1
−3	2	2	1	1	−1
−3	5	1	1	1	1
−2	−5	1	3	2	2
−2	−2	0	2	0	0
−2	1	0	0	0	1
−2	4	0	1	−1	−1
−2	7	−2	0	0	0
−1	−6	2	0	0	0
−1	−3	1	0	0	0
−1	3	−1	−2	−2	0
−1	6	−1	−1	−2	0
−1	9	1	1	1	1
0	−4	−1	−1	0	0
0	2	−2	−2	−2	−2
0	8	1	0	−1	−1
1	−5	2	0	0	0
1	−2	0	1	0	0
1	4	0	−2	−2	−1
1	7	1	1	−1	0
1	10	0	−1	0	0
2	−3	−1	1	1	2
2	0	−2	−2	−2	−2
2	3	0	0	0	1
2	6	0	2	1	1
2	9	0	0	0	−2
3	−4	0	0	0	1
3	−1	0	1	2	2
3	2	1	0	0	0
3	5	−2	−3	−1	−1
3	8	0	−1	0	0
4	−2	2	1	1	0
4	1	2	2	1	0
4	4	0	1	1	1
4	7	1	0	0	0
5	−3	1	1	1	1
5	3	1	3	2	2
5	6	2	0	0	0
6	−1	0	1	1	−1

Continued on the next page

i	j	a_0	a_1	a_2	a_3
6	2	1	1	2	0
6	5	0	2	0	0
7	−2	0	0	0	−2
7	1	0	−1	1	1
7	4	0	0	0	1
8	0	1	0	−1	−1
8	3	0	−1	0	0
9	−1	−1	0	0	0
9	2	−2	0	0	0
10	1	−1	0	0	0

B The central fifth power

Each row $(a, b; h)$ contributes hX^aY^b to $H(X, Y)$. The exponent pairs here are divided by five relative to the output Laurent list in U, V. Thus this table has 75 terms in X, Y and (24) uses $(5a, 5b)$ in the twisted Laurent algebra. All omitted coefficients are zero. Proposition 6.1 and the checked literal-output alignment establish that this is the actual fifth power, not merely a central-looking candidate table.

a	b	h_{ab}
−10	−9	−1
−9	−10	−1
−9	−7	3
−8	−8	14
−8	−5	−6
−7	−9	3
−7	−6	−39
−7	−3	6
−6	−7	−39
−6	−4	84
−6	−1	−3
−5	−8	−6
−5	−5	60
−5	−2	−117
−5	1	−3
−4	−6	84
−4	−3	−90
−4	0	138
−4	3	6
−3	−7	6
−3	−4	−90
−3	−1	21
−3	2	−117

Continued on the next page

a	b	h_{ab}
-3	5	-6
-2	-5	-117
-2	-2	192
-2	1	21
-2	4	84
-2	7	3
-1	-6	-3
-1	-3	21
-1	0	-120
-1	3	-90
-1	6	-39
-1	9	-1
0	-4	138
0	-1	-120
0	2	192
0	5	60
0	8	14
1	-5	-3
1	-2	21
1	1	-120
1	4	-90
1	7	-39
1	10	-1
2	-3	-117
2	0	192
2	3	21
2	6	84
2	9	3
3	-4	6
3	-1	-90
3	2	21
3	5	-117
3	8	-6
4	-2	84
4	1	-90
4	4	138
4	7	6
5	-3	-6
5	0	60
5	3	-117
5	6	-3
6	-1	-39
6	2	84
6	5	-3
7	-2	3

Continued on the next page

a	b	h_{ab}
7	1	-39
7	4	6
8	0	14
8	3	-6
9	-1	-1
9	2	3
10	1	-1

C Reproduction and provenance

The companion file `certificate-data.json` contains exactly the two tables, with integer coefficient quadruples for the numerator and integer coefficients for H. The file `replay-certificate.py` uses only the Python standard library. It implements monic coefficient reduction, twisted Laurent multiplication, the pair construction, exhaustive fifth-power comparison, the two residue-coordinate checks, and the order-three action on Laurent monomials. It is a review aid and external exact-arithmetic check, not a substitute for the Lean kernel.

The companion `source-snapshot.json` records the aggregate source archive hash, source count, final theorem name, and snapshot verification status. It points to the per-source checksum manifest; it does not contain stage-by-stage build receipts. The package verification record and Section 9 distinguish the initial checks, the staged replay status, reused objects, and review caveats.

References

[ABGV11] Asher Auel, Eric Brussel, Skip Garibaldi, and Uzi Vishne. Open problems on central simple algebras. *Transformation Groups*, 16(1):219–264, 2011. Section numbers refer to arXiv:1006.3304v2.

[Fry14] Siân Fryer. The q-division ring and its fixed rings. *Journal of Algebra*, 402:358–378, 2014. Theorem and equation numbers refer to arXiv:1310.5071v2.

[Mat08] Eliyahu Matzri. All dihedral division algebras of degree five are cyclic. *Proceedings of the American Mathematical Society*, 136(6):1925–1931, 2008. Theorem number refers to arXiv:1402.0340.

[RV06] Zinovy Reichstein and Nikolaus Vonessen. Group actions on central simple algebras: a geometric approach. *Journal of Algebra*, 304(2):1160–1192, 2006. Section and proposition numbers refer to arXiv:math/0408420v2.